THE FREE ℓ^1 SEMINORM ON BANACH PRESHEAF COBOUNDARIES

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Abstract. We classify seminorms on the 1-cochain space of a Banach presheaf over a finite poset satisfying additivity, faithfulness, and contractive monotonicity. Specifically, let P be a finite partially ordered set and F a Banach presheaf over P. The *1-cochain space* is $C^1(P,F) = \bigoplus_{e\in E_{\mathrm{cov}}} F(a_e)$, where the direct sum is taken in the ℓ^1 sense. We prove that the ℓ^1 coboundary norm $I(s) = \sum_e \|\delta_e(s)\|$ is, up to a positive scalar, the **unique** seminorm on $C^1(P,F)$ satisfying these three properties. Adding isomorphism invariance forces the scalar to be uniform across edges. No choice is involved; the norm is generated by the constraints.

In short: *any additive, faithful, contractively monotone diagnostic on Banach presheaf coboundaries must be a positive scalar multiple of the ℓ^1 norm.*

1. Setup

1.1. **Banach Presheaves over Finite Posets.** Let $(P,\leq)$ be a finite partially ordered set. The *covering relations* are the edges of the Hasse diagram:

$$E_{\mathrm{cov}}(P) = \{e = (a,b) : a < b,\ \nexists\, c \text{ with } a < c < b\}.$$

A *Banach presheaf* F over P assigns:

- To each $x \in P$, a Banach space $(F(x), \|\cdot\|_{F(x)})$.
- To each $e = (a,b) \in E_{\mathrm{cov}}$, a bounded linear *restriction map* $r_{b,a} : F(b) \to F(a)$ with $\|r_{b,a}\|_{\mathrm{op}} \leq 1$.

A *section* of F is a family $s = (s(x))_{x\in P}$ with $s(x) \in F(x)$.

1.2. **The 1-Cochain Space.** For each edge $e = (a,b)$, the *coboundary* is:

$$\delta_e(s) = r_{b,a}(s(b)) - s(a) \in F(a).$$

The *1-cochain space* is the Banach direct sum:

$$C^1(P,F) = \bigoplus_{e\in E_{\mathrm{cov}}}^{\ell^1} F(a_e)$$

with norm $\|(\delta_e)_e\|_1 = \sum_e \|\delta_e\|_{F(a_e)}$. This is isometrically isomorphic to the space of coboundary vectors.

The ℓ^1 *coboundary norm* of a section s is:

$$I(s) = \sum_{e\in E_{\mathrm{cov}}} \|\delta_e(s)\|_{F(a_e)} = \|(\delta_e(s))_e\|_1.$$

1.3. **Seminorms on 1-Cochains.**

Definition 1.1. A *seminorm on 1-cochains* is a functional $\nu : C^1(P,F) \to \mathbb{R}_{\geq 0}$.

We consider four properties:

(P1) Edge-wise additivity. ν decomposes as

$$\nu((\delta_e)_e) = \sum_{e\in E_{\mathrm{cov}}} \nu_e(\delta_e)$$

where each $\nu_e : F(a_e) \to \mathbb{R}_{\geq 0}$ is itself a seminorm on $F(a_e)$.

(P2) Faithfulness. For each e: $\nu_e(\delta) = 0 \iff \delta = 0$.

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(P3) Contractive monotonicity. For every bounded linear map $\varphi : V \to W$ between Banach spaces, and every presheaf morphism induced stalkwise by φ:

$$\nu_e(\varphi(\delta)) \leq \|\varphi\|_{\mathrm{op}} \cdot \nu_e(\delta).$$

(P4) Isomorphism invariance. For any poset isomorphism $\pi : \tilde{P} \to P$:

$$\nu_{\tilde{P}}(\pi^*(\delta_e)_e) = \nu_P((\delta_e)_e).$$

2. The Classification Theorem

We first isolate two structural lemmas.

Lemma 2.1 (1-Dimensional Compatibility)**.** *Let ν_e satisfy* (P3)*. Then for any 1-dimensional Banach subspace* $\mathrm{span}(\delta) \subset V$*, the restriction $\nu_e|_{\mathrm{span}(\delta)}$ is well-defined and satisfies* (P1)–(P3) *with respect to the induced Banach norm on* $\mathrm{span}(\delta) \cong \mathbb{R}$*. Specifically: any isometric embedding $\iota : \mathbb{R} \hookrightarrow V$ induces a seminorm $\nu_e^{\mathbb{R}} = \nu_e \circ \iota$ on $\mathbb{R}$ satisfying* (P2)–(P3)*, and any contractive surjection $f : V \to \mathbb{R}$ induces $\nu_e^{\mathbb{R}}(f(\delta)) \leq \|f\|_{\mathrm{op}} \cdot \nu_e(\delta)$.*

Proof. Direct application of (P3) to ι and f respectively. Both are bounded linear maps between Banach spaces, so (P3) applies without additional hypothesis. □

Lemma 2.2 (Direction Independence)**.** *If ν_e satisfies* (P3)*, then for any $v, w \in F(a_e)$ with $\|v\| = \|w\| \neq 0$, we have $\nu_e(v) = \nu_e(w)$.*

Proof. By the Hahn–Banach theorem, there exists $f_v \in F(a_e)^*$ with $\|f_v\|_{\mathrm{op}} = 1$ and $f_v(v) = \|v\|$. Define $\Phi : F(a_e) \to F(a_e)$ by $\Phi(x) = f_v(x)\frac{w}{\|w\|}$. Then $\Phi(v) = w$ and $\|\Phi\|_{\mathrm{op}} = 1$, so $\nu_e(w) \leq \nu_e(v)$ by (P3). By symmetry, constructing $\Psi(x) = f_w(x)\frac{v}{\|v\|}$ gives $\nu_e(v) \leq \nu_e(w)$, forcing equality. □

Theorem 2.3 (Free Seminorm Classification)**.** *Let ν be a seminorm on $C^1(P, F)$ satisfying* (P1)*,* (P2)*, and* (P3)*. Then for each edge e, there exists $w_e > 0$ such that*

$$\nu_e(\delta) = w_e \cdot \|\delta\|_{F(a_e)} \quad \textit{for all } \delta \in F(a_e).$$

If additionally (P4) *holds, then $w_e = \alpha$ is independent of e, and*

$$\nu((\delta_e)_e) = \alpha \sum_e \|\delta_e\|_{F(a_e)} = \alpha \cdot I(s).$$

In other words: the ℓ^1 coboundary norm is the unique (up to positive scalar) seminorm satisfying all four properties.

3. Proof

The proof proceeds in four stages: direction independence, factoring through norms, forcing linearity, and forcing uniformity.

Stage 1: Direction Independence. This is Lemma 2.2 above.

Stage 2: Each ν_e depends only on $\|\delta\|$. Fix an edge e and let $V = F(a_e)$. Fix $\delta \in V$ with $\|\delta\| = r > 0$.

Upper bound. By the Hahn–Banach theorem, there exists $f \in V^*$ with $\|f\|_{\mathrm{op}} = 1$ and $f(\delta) = r$. View f as a bounded linear map $f : V \to \mathbb{R}$ with $\|f\|_{\mathrm{op}} = 1$. By Lemma 2.1 and (P3):

$$\nu_e^{\mathbb{R}}(f(\delta)) \leq \|f\|_{\mathrm{op}} \cdot \nu_e(\delta) = \nu_e(\delta)$$

where $\nu_e^{\mathbb{R}} := \nu_e \circ \iota$ for any unit-norm isometric embedding $\iota : \mathbb{R} \hookrightarrow V$. By Lemma 2.2, two embeddings mapping 1 to vectors of equal norm produce identical values of ν_e, so $\nu_e^{\mathbb{R}}$ is well defined. Since $f(\delta) = r$, we get $\nu_e^{\mathbb{R}}(r) \leq \nu_e(\delta)$.

Lower bound. Define $\iota : \mathbb{R} \to V$ by $\iota(t) = t \cdot \delta/\|\delta\|$. Then $\|\iota\|_{\mathrm{op}} = 1$ (it maps t to a vector of norm $|t|$). By (P3) applied to ι:

$$\nu_e(\iota(r)) \leq \|\iota\|_{\mathrm{op}} \cdot \nu_e^{\mathbb{R}}(r) = \nu_e^{\mathbb{R}}(r).$$

Since $\iota(r) = \delta$, we have $\nu_e(\delta) \leq \nu_e^{\mathbb{R}}(r)$.

Combining: $\nu_e(\delta) = \nu_e^{\mathbb{R}}(\|\delta\|)$.

For $\delta = 0$: $\nu_e(0) = 0$ by (P2). Therefore $\nu_e(\delta) = h_e(\|\delta\|)$ where $h_e : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ with $h_e(0) = 0$ and $h_e(r) > 0$ for $r > 0$. □ (Stage 2)

Stage 3: h_e is linear. Apply (P3) to the scaling maps $v \mapsto cv$ and $v \mapsto v/c$ for $c > 0$, with operator norms c and $1/c$ respectively. These yield:

$$h_e(cr) \leq c \cdot h_e(r) \quad \text{and} \quad c \cdot h_e(r) \leq h_e(cr).$$

Together: $h_e(c \cdot r) = c \cdot h_e(r)$ for all $c > 0$ and $r \geq 0$.

This is exact 1-homogeneity. Setting $r = 1$:

$$h_e(c) = c \cdot h_e(1) \quad \text{for all } c > 0.$$

Combined with $h_e(0) = 0$ (faithfulness), this gives $h_e(d) = w_e \cdot d$ on all of $\mathbb{R}_{\geq 0}$, where $w_e = h_e(1) > 0$ (positive by (P2)). □ (Stage 3)

Stage 4: Isomorphism invariance forces uniform weights. Let $e = (a, b)$ be any covering edge of a finite poset P. Define the *single-edge subposet* $P_e = \{a < b\}$ with the restricted presheaf $F_e = F|_{P_e}$, and let $\iota_e : P_e \hookrightarrow P$ denote the inclusion functor.

By (P1), ν decomposes as $\nu((\delta_e)_e) = \sum_{e'} \nu_{e'}(\delta_{e'})$. Each summand $\nu_{e'}$ depends only on $\delta_{e'} \in F(a_{e'})$ and is determined independently of the other summands. In particular, $\nu_e(\delta_e)$ depends only on the coboundary $\delta_e(s) = r_{b,a}(s(b)) - s(a)$, which is computed entirely from the restricted presheaf data $(F(a), F(b), r_{b,a})$ — that is, from F_e alone. The value $\nu_e(\delta_e)$ is therefore independent of the ambient poset P in which e is embedded: it is determined by the single-edge presheaf (P_e, F_e).

By Stages 1–3, $\nu_e(\delta_e) = w_e\|\delta_e\|$ for some $w_e > 0$. Since w_e depends only on the isomorphism class of the single-edge presheaf (P_e, F_e), it remains to show all such presheaves yield the same weight.

Let $e_1 = (a_1, b_1)$ and $e_2 = (a_2, b_2)$ be any two covering edges (in the same or different posets) with the same stalk spaces $F(a_{e_1}) \cong F(a_{e_2})$ and $F(b_{e_1}) \cong F(b_{e_2})$. The single-edge posets $P_{e_1} = \{a_1 < b_1\}$ and $P_{e_2} = \{a_2 < b_2\}$ admit a unique order-preserving bijection $\pi : P_{e_1} \to P_{e_2}$. By (P4), $\nu_{P_{e_1}}(\pi^*(\delta_{e_2})) = \nu_{P_{e_2}}((\delta_{e_2}))$, which gives $w_{e_1} = w_{e_2}$.

Thus all covering edges share the same weight w.

Setting $\alpha = w$: $\nu = \alpha \cdot I$. □ (Theorem 2.3)

4. Necessity of Each Property

Dropped property	What happens	Example
(P1) Additivity	ℓ^∞-type norms admitted	$\nu = \max_e \|\delta_e\|$
(P2) Faithfulness	Trivial seminorm admitted	$\nu \equiv 0$
(P3) Contractivity	Nonlinear seminorms admitted	$\nu = \sum_e \|\delta_e\|^2$
(P4) Invariance	Non-uniform weights admitted	$\nu = \sum_e w_e\|\delta_e\|$

Each property is independently necessary. Together they are sufficient.

Remark 4.1. Properties (P1)–(P3) alone force the ℓ^1-type structure $\nu = \sum_e w_e\|\delta_e\|$ with possibly non-uniform positive weights. Property (P4) collapses the weights to a single scalar.

5. The Universal Property (Categorical Formulation)

Definition 5.1. Let **AFC** (Additive Faithful Contractive) denote the category whose:

- **Objects** are seminorms ν on $C^1(P, F)$ satisfying (P1)–(P3) for all finite posets P and Banach presheaves F.
- **Morphisms** $\eta : \nu \to \mu$ are given by families of positive reals $\{\lambda_e\}$ with $\mu_e = \lambda_e \cdot \nu_e$ for all e.

This category is not necessarily small as it spans all Banach presheaves and posets. Furthermore, it is a thin category with respect to any pair of objects, as the morphisms are simply determined by structural scalars relating the seminorms.

Corollary 5.2 (Structure of AFC)**.** *Every object in* **AFC** *has the form* $\nu = \sum_e w_e \|\cdot\|_{F(a_e)}$ *for some family* $\{w_e > 0\}$*. In particular, every object in* **AFC** *is a weighted* ℓ^1 *norm.*

Corollary 5.3 (Universal Property)**.** *Let* $\mathbf{AFC}_4 \subset \mathbf{AFC}$ *denote the full subcategory of objects also satisfying* (P4)*. Then:*

(i) Every object in $\mathbf{AFC}_4$ *equals* $\alpha \cdot I$ *for some* $\alpha > 0$*.*

(ii) I *is terminal in* $\mathbf{AFC}_4$ *up to positive scalar normalization: for each* $\nu = \alpha \cdot I$*, the unique morphism* $\nu \to I$ *is the global scalar* $1/\alpha$*.*

(iii) The universal property diagram commutes: for any $\nu \in \mathbf{AFC}_4$ *and any Banach presheaf morphism* $\varphi : F \to G$*, the square*

$$\begin{array}{ccc} C^1(P,F) & \xrightarrow{\varphi_*} & C^1(P,G) \\ \downarrow \nu & & \downarrow \nu \\ \mathbb{R}_{\geq 0} & = & \mathbb{R}_{\geq 0} \end{array}$$

satisfies $\nu(\varphi_*(\delta)) \leq \|\varphi\|_{\mathrm{op}} \cdot \nu(\delta)$*, and* I *is the unique (up to scalar) seminorm for which this holds.*

6. Discussion

6.1. **What This Paper Proves.** The ℓ^1 coboundary norm is not selected from a family of candidates. It is *generated* by three structural constraints (additivity, faithfulness, contractivity) on the space of 1-cochains of Banach presheaves over finite posets. Adding relabeling invariance eliminates the only remaining freedom (edge weights). The result is purely functional-analytic — it requires no physics, no dynamics, and no topology beyond the Hasse diagram of a finite poset.

6.2. **What This Paper Does Not Prove.**

- We do not prove anything about dynamics, time evolution, or physical systems.
- We do not extend to infinite posets (where compactness arguments would be needed).
- We do not claim the four properties are the "right" axioms. We prove that *if* they hold, the norm is forced.

6.3. **Relation to Other Work.** The Hahn–Banach factoring argument used in Stage 2 is analogous to classical norm classification techniques in Banach space theory, where linear functionals are used to reduce geometric constraints to one-dimensional analysis (see Megginson [3] for Banach geometric tools, and Pestov [1] for related uniqueness results for norms). The key structural observation is that contractive monotonicity, combined with Hahn–Banach, forces exact 1-homogeneity — eliminating all nonlinear seminorms without any regularity assumption. This mechanism parallels the uniqueness proof for the operator norm, where submultiplicativity and compatibility with vector norms force norm-dependence via analogous Hahn–Banach arguments. The scaling argument in Stage 3 is elementary but, to our knowledge, has not been applied in this presheaf context (though related categorical sheaf structures appear in Godement [5]).

The result is a presheaf version of a classical principle in Banach space theory: the characterization of the ℓ^1 norm as the unique additive cross-norm on a free Banach sum. The 1-cochain space $C^1(P,F) = \bigoplus_e F(a_e)$ is the *free additive Banach object generated by the edges* of the poset. When a measurement on such a free object must satisfy additivity across components and compatibility (via contractive monotonicity) with the norms on each component, ℓ^1 is the only geometry that survives. The ℓ^2 norm fails because $\|(v,w)\|_2 \neq \|v\| + \|w\|$; the ℓ^∞ norm fails because $\max(\|v\|, \|w\|)$ discards summation. This is why ℓ^1 appears so inevitably: it is the *universal additive geometry.*

We note that the coboundary defects $\delta_e(s) = r_{b,a}(s(b)) - s(a)$ play a role analogous to curvature in gauge theories: they measure the failure of local data to glue consistently across the covering relations of the poset. In gauge theory, curvature measures the non-commutativity of

parallel transport around loops; here, the coboundary measures the incompatibility of section values across edges. The distinction is that in gauge theory the metric on curvature is a free choice, whereas our classification theorem shows that the defect magnitude is uniquely ℓ^1 — the geometry forces the norm.

6.4. **Relation to the Main SOT Paper.** This paper serves as a companion to the Single Obstruction Theory. Together, the two papers form a classical three-layer structure:

(1) **Existence** (SOT Theorem 3.1): Nonlinear projection introduces curvature — a differential-geometric obstruction measured by the second-order term $D^2\Pi(Am, v)$, which acts as a second fundamental form.
(2) **Uniqueness** (this paper, Theorem 2.3): The obstruction must be measured by the ℓ^1 coboundary norm. No other norm is compatible with the structural constraints.
(3) **Persistence** (SOT Theorem 5.1): The obstruction density $C(t) \geq c > 0$ persists at finite density for physical time intervals independent of system size.

The main SOT paper asserts ℓ^1 rigidity as a structural input; this paper proves *why* ℓ^1 is forced. The separation is deliberate: the classification theorem is a self-contained result in functional analysis that neither requires nor implies any dynamical or physical interpretation. It answers the question: *why* ℓ^1 *and not* ℓ^2*?* Because the axioms force it.

References

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